

## End-to-End Guide for My SQL Project – FooDrive

This guide walks through everything I did to build a **SQL project based on FooDrive**, a fictional food delivery company. I've broken down each step from **setting up the database** to **solving real-world business problems using SQL**. This helped me practice **real-time data handling** and **analytical thinking**.

### Step 1: Setting Up PostgreSQL and pgAdmin 4

#### 1. Installing PostgreSQL:

I downloaded and installed PostgreSQL from the official website. During setup, I set a password for the default postgres user (you'll need it later).

#### 2. Installing pgAdmin 4:

I installed pgAdmin 4 — a **graphical interface** that simplifies managing PostgreSQL databases.

#### 3. Launching pgAdmin 4:

After installation, I launched pgAdmin 4 and **logged in using the password** I created during setup.

### Step 2: Creating the Database

#### 1. Setting Up My Database:

Inside pgAdmin 4, I right-clicked on **"Databases"** → **"Create"** and made a new database called FooDrive\_db. The owner was kept as postgres.

### Step 3: Creating Tables

#### 1. Building the Tables:

Under **Schemas** → **Tables**, I created the core tables: restaurants, customers, orders, riders, and deliveries.

I carefully defined:

- **Data types**
- **Primary and foreign keys**
- **Constraints** to maintain data integrity

### Step 4: Importing Data

#### 1. Getting the Data Ready:

I ensured that my **CSV files matched the structure** of my tables — column names, formats, and datatypes.

#### 2. Importing in pgAdmin:

Using **Import/Export** in pgAdmin, I:

- Selected the CSV file
- Mapped the columns correctly
- Ran the import

This loaded all my data into the tables.

## Step 5: Checking for Null Values

### 1. Finding Missing Data:

I wrote **SQL queries** to check for null values in important columns.

### 2. Handling Nulls:

Depending on the context, I:

- **Removed rows**
- **Filled in missing values**
- **Flagged records** for manual review

## Step 6: Doing Exploratory Data Analysis (EDA)

### 1. Getting to Know the Data:

I ran **basic SELECT queries** to explore:

- Record counts
- Unique values
- Table relationships

### 2. Visualizing the Data:

I used:

- **pgAdmin options**
- **Excel or Python** (for bar charts, histograms, etc.)

This helped identify **trends and outliers** visually.

## Step 7: Basic Data Analysis

### 1. Running Basic Queries:

I started with simple analysis like:

- Total orders
- Average order value
- Top restaurants/customers

### 2. Summarizing Key Insights: I

focused on key metrics like:

- **Total revenue**
- **Active users**
- **Order frequency**

These insights helped evaluate FooDrive's **business performance**.

## Step 8: Solving the 20 Business Questions

### 1. Understanding the Questions:

I used a curated list of **20 business-focused SQL problems** to simulate real-world scenarios.

**2. Breaking Down Each Problem:** For each question, I:

- Identified relevant tables
- Wrote the query step-by-step
- Tested and validated logic

**3. Saving My Work:**

I documented all my **SQL queries** in a well-organized format.

**4. Presenting the Results:**

I compiled all outputs to prepare a **dashboard or report**, making it look professional for sharing.

---